Base board



File: base.kicad_sch

dial-me-spinner rev A — generated by hardware/design/build.py from hardware/design/sheets/*.py.
Do not hand-edit the .kicad_sch files: change the sheet modules and rebuild.
Architecture of record: hardware/design/SPEC.md

dial-me-spinner

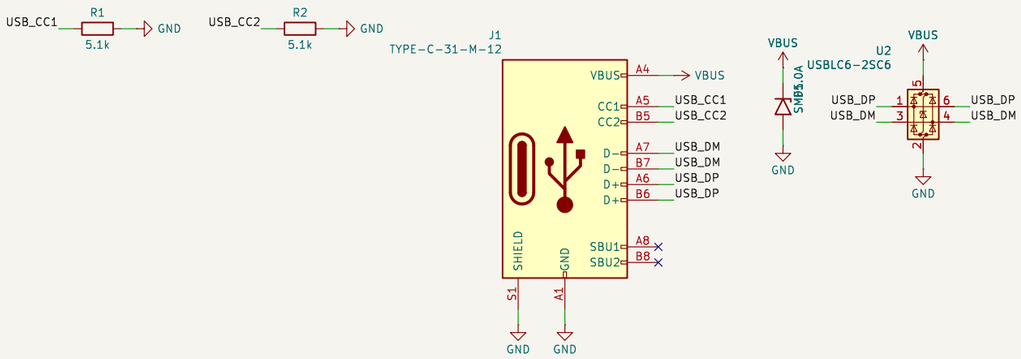
Sheet: /
File: dial-me-spinner-base.kicad_sch

Title: dial-me-spinner base board

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/2

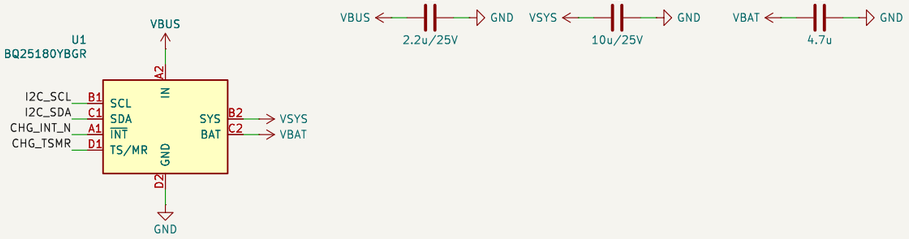
USB-C input and protection

HRO TYPE-C-31-M-12 (sink, USB 2.0). Separate 5.1k Rd per CC pin; D+/D- pairs tied at the connector.
USBLC6-2SC6 and SMF5.0A within 5 mm of J1; route D+/D- through the ESD pads (flow-through) as a 90R pair.
J1 body (3.26 mm tall, 7.35 mm deep) reaches r ~27.6, under the FA lower plate (r <= 30): plate needs a relief there.



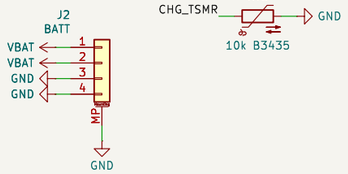
Charger and power path (BQ25180, I2C 0x6A)

IN 2.2u/25V, SYS 10u/25V, BAT 4.7u: each within 1 mm of its ball and of the D2 GND ball.
SYS = VSYS (4.4 V on USB, SYS_REG set by firmware; VBAT on battery). /INT, SCL, SDA pull-ups are on the main board.



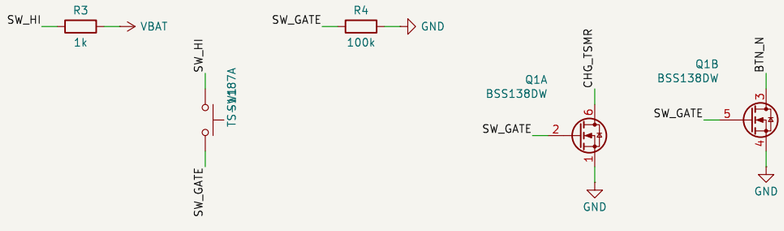
Battery connector J2 and NTC RT1 (TS/MR)

J2 SM04B side-entry SH (2.95 mm): pins 1-2 = BAT+, 3-4 = GND (two 28 AWG leads per rail). Packs are re-terminated: check the polarity of every pack.
RT1 on TS/MR: top side, at the board edge farthest from U1 (>= 10 mm), no copper shared with U1. Bottom side has no parts, so RT1 reads board temperature, not the cell. GND Kelvin to D2.
No cap on TS/MR (TS bias is pulsed 4 ms / 196 ms).



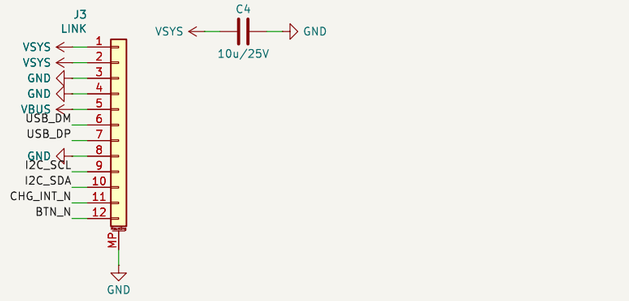
Centre press switch and ship-mode wake

SW1 at the board centre (top). Press: VBAT -> 1k -> SW1 -> SW_GATE (100k to GND, ~42 uA while held; the gates discharge within the BQ25180 deglitch on release).
Q1A pulls TS/MR low (ship wake / 20 s HW reset); Q1B pulls BTN_N low (nRF SENSE wake). BTN_N pull-up is on main.



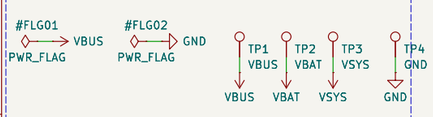
Board-to-board link (to main board through the spindle)

JST SH 12-way SIDE-entry (SM12B, 2.95 mm) at r 15.5, θ 120°, entry facing the centre; identical pinout on the main board (SPEC §2). C4 sits next to J3. The harness runs into the spindle's side slot.



Test points and power flags

PWR_FLAGs mark VBUS (USB-C) and GND as externally driven for ERC.



dial-me-spinner		
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Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 2/2